ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

Pixel pitch = 0.005 mm, sensor width = 1280x0.005 = 6.4 mm. how to calculate the belt field of view.

Q1(b) [3 marks]

ADC step = 5/2^12 = 0.0012207 V = 1.2207 mV. Sensor sensitivity is 10 mV/N, so ΔF = 1.2207/10 = 0.12207 N. The smallest resolvable force change is approximately 0.122 N.

Calculation: $2^{12} = 4096$ levels. One step is $5/4096 = 0.0012207$ V, i.e. 1.2207 mV. Dividing by the sensitivity, $1.2207\,\text{mV}/(10\,\text{mV/N}) = 0.12207$ N. The mV units cancel, leaving N.

Q1(c) [3 marks]

$$F = e^(- 2 \times 0.1) = 0.818731$$

I used G = (1-F)/(-2) = -0.0906346. So:

$$x_{k+1} = 0.818731 x_k - 0.0906346 u_k$$

Question 2 [10 marks]

Q2(a) [2 marks]

$\dot{x}_1 = x_2$ and $\dot{x}_2 = (\tau - b\, x_2 + mg\ell \sin\theta)/J$. At equilibrium:

$$_0 = -mg\ell sin\theta_e,$$

an equilibrium exists for $|\tau_0|$ less than or equal to 2.4525 N m

Q2(b) [2 marks]

With $x = [\theta, \dot{\theta}]^T$, A = [[0,1], [-(mgℓ/J)cos θ_e, -b/J]], B = [[0],[1/J]]. At θ_e = 60°, A = [[0,1], [-61.3125, -2.5]] and B = [[0],[50]].

At the equilibrium, the constant torque balances gravity. For a small perturbation, use $\sin(\theta_e + \delta\theta) \approx \sin\theta_e + \cos\theta_e\,\delta\theta$. The constant terms cancel against τ_0; the remaining linear term gives the local state matrix.

Q2(c) [4 marks]

At rest, $\dot{\theta} = \ddot{\theta} = 0$, so the torque balance is $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$. With $K_p = 50$, $\theta_d = \pi/2$ and $mg\ell = 2.4525$, solve $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$. This gives θ_{ss} approximately 1.52181 rad = 87.193°, so the error is 90°-87.193° = 2.807°; it is not zero.

Rearranging the steady-state equation gives $50(\pi/2 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$. I used radians in the equation. Solving numerically gives $\theta_{ss} \approx 1.522$ rad $\approx 87.19°$, so the gap from $90°$ is about $2.81°$.

Q2(d) [2 marks]

With gravity compensation, the gravity term cancels the plant's $mg\ell \sin\theta$ term. At rest, $K_p(\theta_d - \theta_{ss})=0$, so $\theta_{ss}=\theta_d$ and the steady-state error is zero. The linearised gravity stiffness term is also cancelled, so the closed-loop poles generally change; the stiffness contribution $mg\ell \cos\theta_d$ is removed.

The compensation term cancels the nominal gravity torque in the plant model. At steady state the proportional term must then be zero, which requires $\theta_d - \theta_{ss} = 0$. This does not mean the transient response is unchanged: removing the gravity derivative changes the linearized stiffness.

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) [2 marks]

I use ω_c=20 rad/s and $\alpha=\omega_c T/2$=0.01. Then:

$$y[k] = 0.980198y[k-1] + 0.009901(x_k + x[k-1])$$

The DC gain is approximately 1.

Q3(c) [4 marks]

The characteristic polynomial is $s^2+(\ell_1+2)s+(2\ell_1+\ell_2)$. I match it to $s^2+10s+100$ and get $\ell_1=8$, $\ell_2=84$.

Q3(d) [2 marks]

The observer is simpler because it only smooths the measured velocity. The filter is safer when the model is poor.

Question 4 [10 marks]

Q4(a) [2 marks]

The closed-loop poles are -2±2j. I use Nff=c/b=6 for unity DC gain.

Q4(b) **[2 marks]**

$$\theta_{ss} = 0.75 \times 8/(6+2) = 0.75,$$

the percentage tracking error is 25%

Q4(c) **[4 marks]**

The characteristic polynomial is $s^3+(a+bk_2)s^2+(c+bk_1)s+bK_I$. I use the target $s^3+6s^2+12s+16$, giving $k_2=4$, $k_1=6$, $K_I=16$.

Q4(d) **[2 marks]**

Integral action gives zero steady-state error if stable. A practical problem is integrator windup.